

How To Ultracold Atom-Ion Experiment

Madison group seminar talk

Albert-Ludwigs-Universität Freiburg

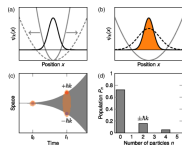
UNI
FREIBURG

Fabian Thielemann, Joachim Welz, Wei Wu, Daniel Hönig, Thomas Walker
and Tobias Schätz
Physikalisches Institut
24. November 2021

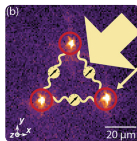
Complexity



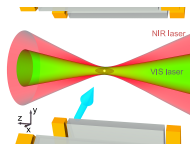
Paula



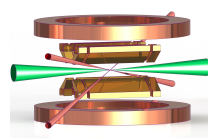
Bermuda



Tiamo



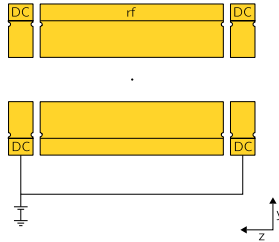
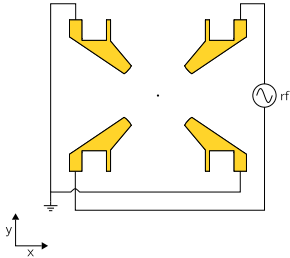
Bali



Control

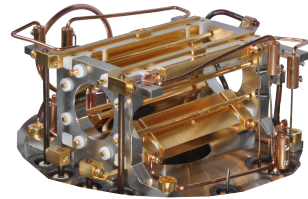
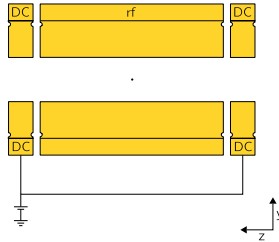
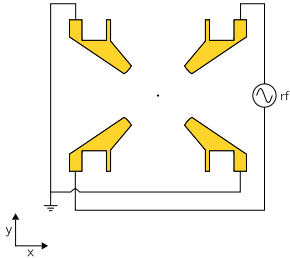
Ion Trapping in RF Fields

Poisson's equation forbids 3D DC confinement: $\Delta\phi_{\text{el}} = 0$
 $\Rightarrow$ Apply time-varying potential in x-y-plane



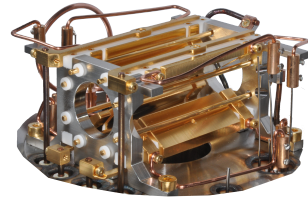
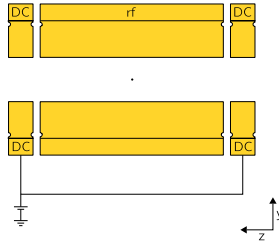
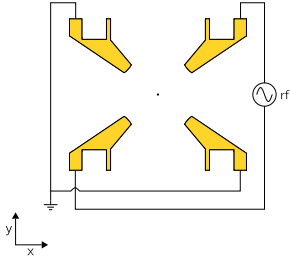
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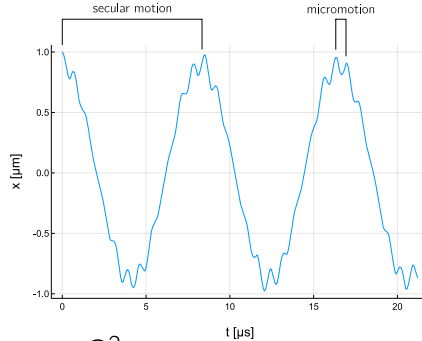
Dynamics of trapped ion are governed by Mathieu equation

$$\ddot{r}_i + [a_i + 2q_i \cos(\Omega_{\text{rf}} t)] \frac{\Omega_{\text{rf}}^2}{4} r_i = 0$$

Ion Trapping in RF Fields

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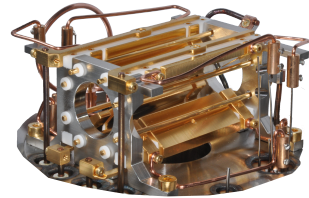
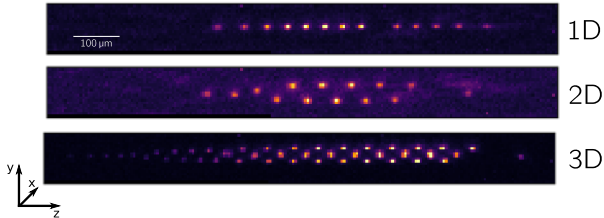
⇒ Apply time-varying potential in x-y-plane



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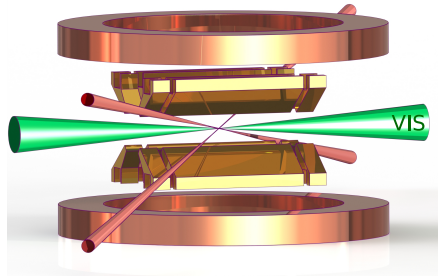
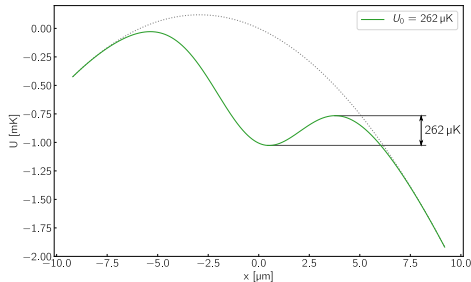
Poisson's equation forbids 3D DC confinement: $\Delta\phi_{\text{el}} = 0$
 $\Rightarrow$ Apply time-varying potential in x-y-plane



Deep potential ($\mathcal{O}(10^3 \text{ K})$) allows storing ion coulomb crystals for hours.

(But we only want single ions and throw them away every few minutes.)

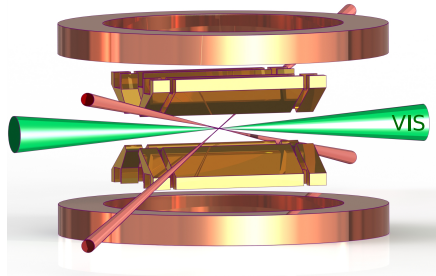
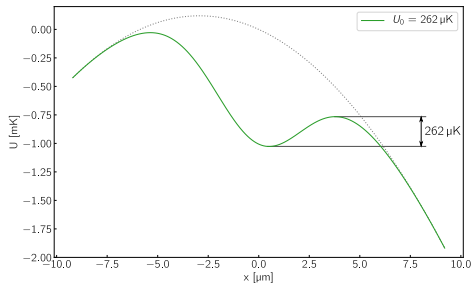
No micromotion: HOORAY! (But also it's much more difficult.)



Ion is sensitive to stray electric fields E_{stray} and DC defocussing

$$U_{\text{tot}}(r) \approx U_{\text{opt}}(r) - eE_{\text{stray}} r - \frac{1}{2} \omega_{\text{dc-defoc}}^2 m_{\text{ion}} r^2$$

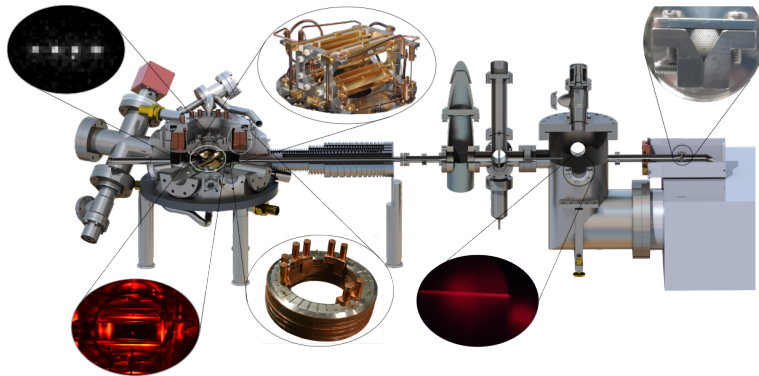
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Shallow, finite potential allows to probe radial ion kinetic energy!

The BaLi Experiment

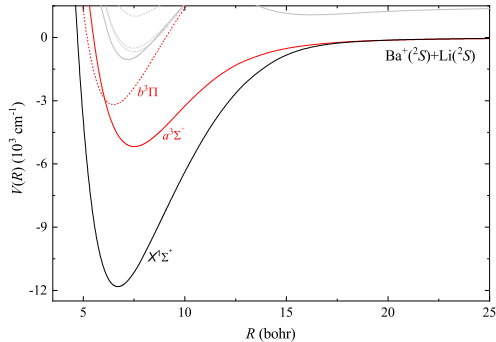
An apparatus to combine single (to tens) of Doppler cooled Ba^+ -ions
and $\approx 2 \cdot 10^4$ ^6Li -atoms at $T \approx 1\mu\text{K}$



Atom-Ion Experiments

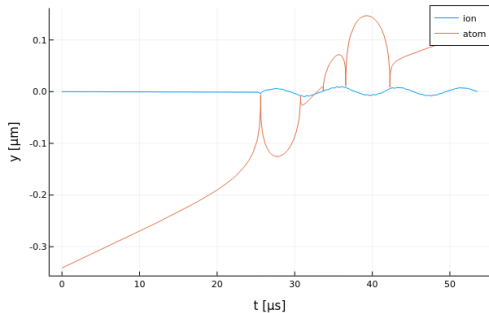
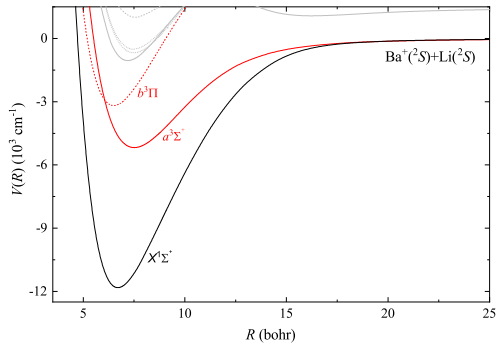
The ion induces a dipole moment in the atom ($p \propto \alpha E(r) \propto \frac{1}{r^2}$) that leads to charge-dipole interaction ($\propto \frac{p}{r^2}$)

$$V_{\text{atom-ion}}(r) = \frac{C_6}{r^6} - \frac{C_4}{r^4}$$



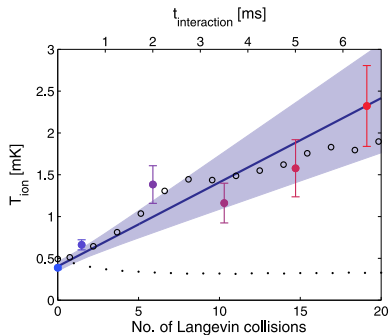
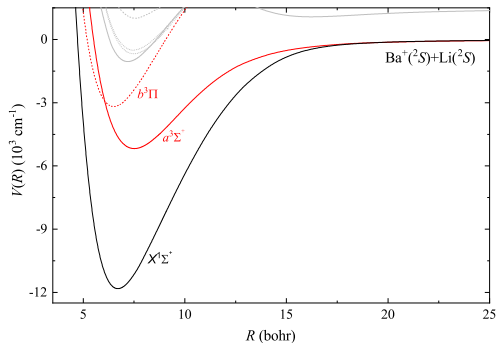
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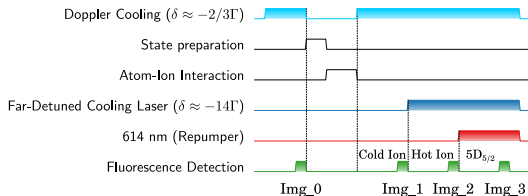
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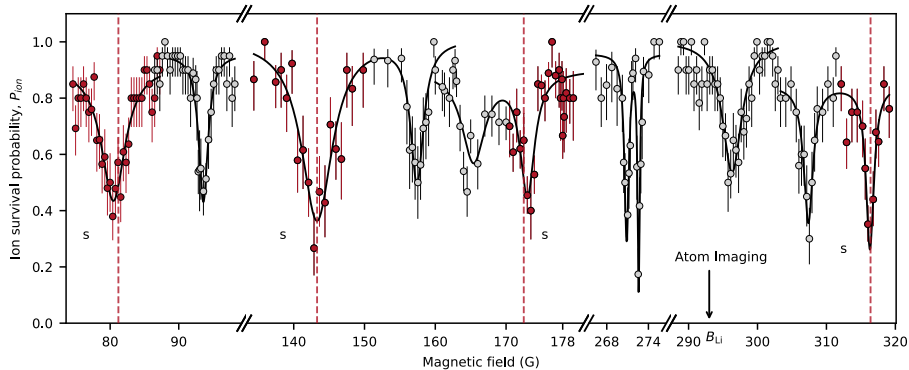
Ziv Meir, et al. "Dynamics of a ground-state cooled ion colliding with ultracold atoms." *Physical review letters* 117.24 (2016): 243401.

We interrogate the ion product state and classify it in four different events.



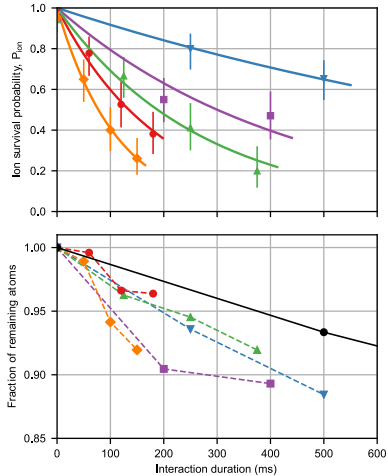
Classification	Img_0	Img_1	Img_2	Img_3
Ion Survival				
Ion Loss				

Feshbach Resonances: Spectrum for Li in $|2\rangle$



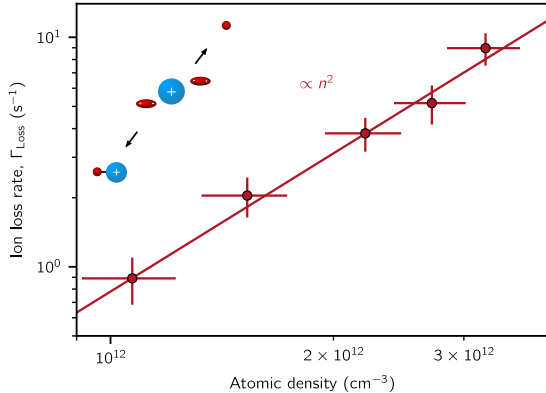
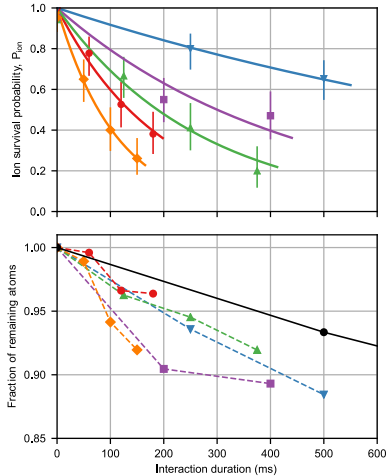
Free fit of a_S and a_T including coupling to s , p and d assigns four s -wave resonances!
(Collaboration with M. Tomza and K. Jachymski in Warsaw)

Density Dependence of Ba^+ Loss on Resonance



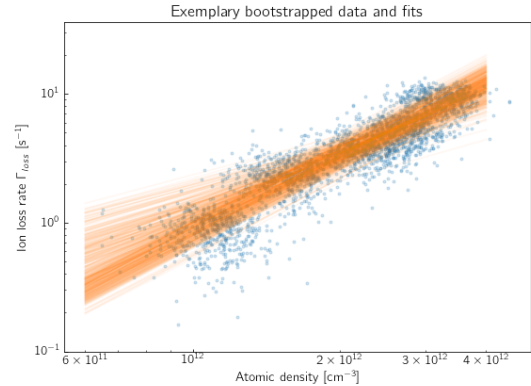
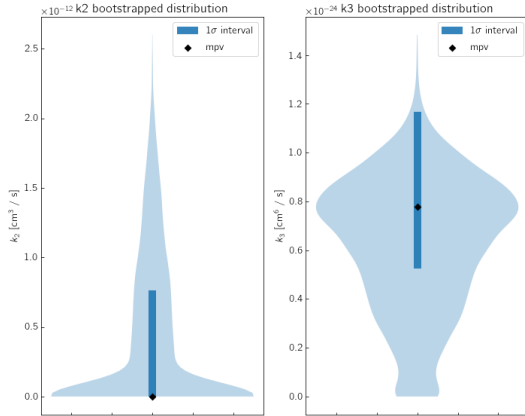
Vary atom number and probe ion decay rate.

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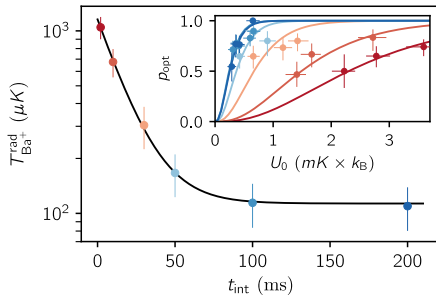
Vary atom number and probe ion decay rate.
 $\Rightarrow$ TBR is dominant loss process on resonance!

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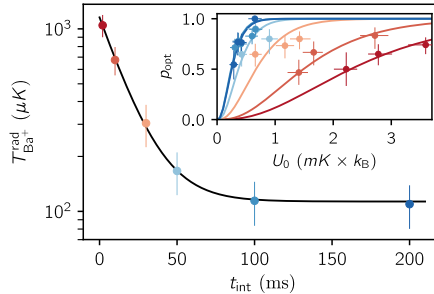


⇒ TBR is dominant loss process on resonance!

Enhanced Cooling of Ba^+ on Resonance

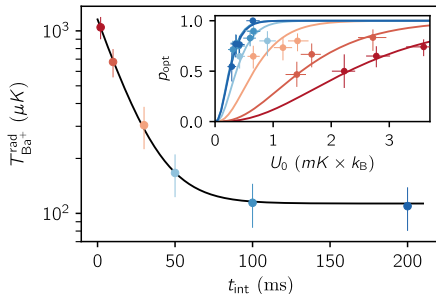
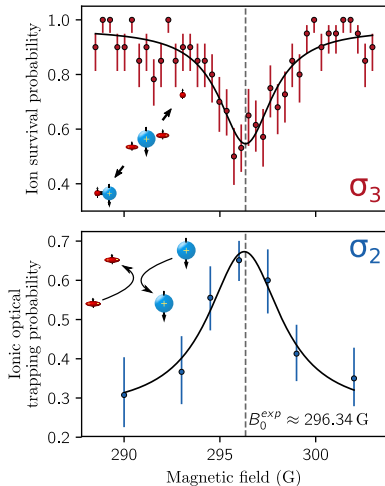


Enhanced Cooling of Ba^+ on Resonance



Prepare a low density atomic cloud to suppress TBR and probe ion radial kinetic energy before full thermalization

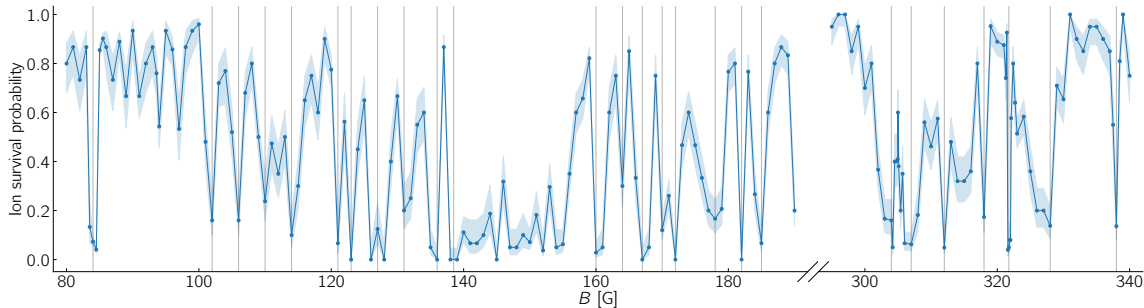
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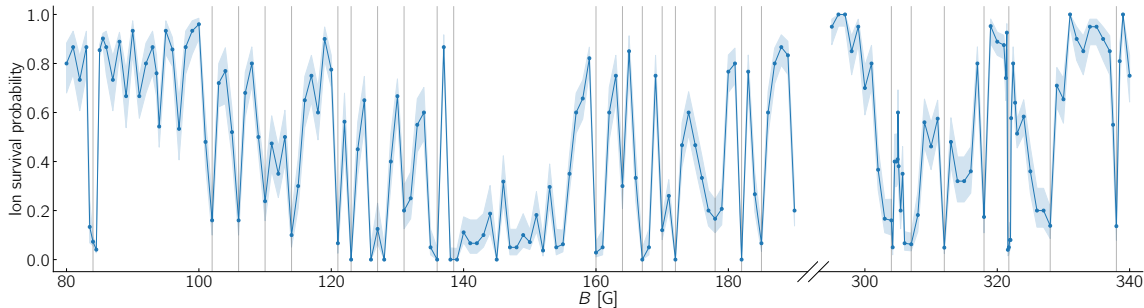
The Forest of Resonances for Li in $|1\rangle$

We scanned a range of 160 G and didn't find a region with flat background > 5 G wide.



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Some resonances are much narrower than what we observed in the other channel, theory explores higher order spin-orbit coupling.

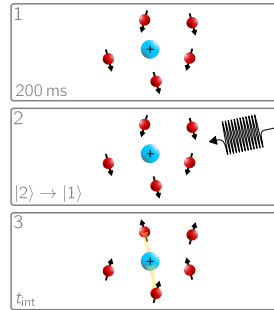
Effect of Collisional Spin Polarization of Ba^+ (?)

$B = 321.75 \text{ G}$ which is on a narrow resonance for Li in $|1\rangle$ but background in $|2\rangle$.

Collisional spin polarization is expected to only work with Li in $|2\rangle$!

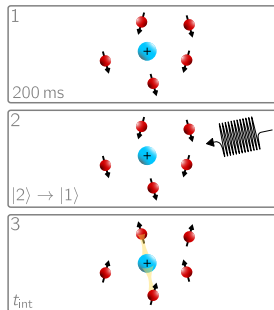
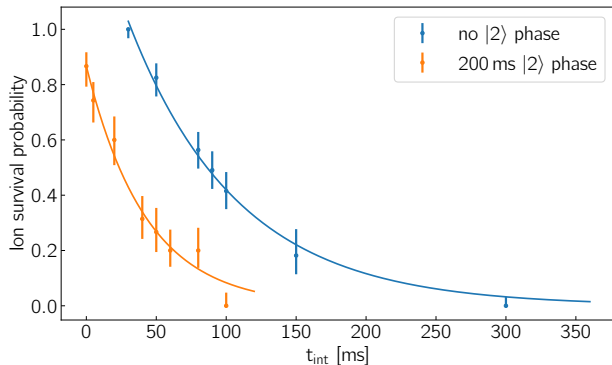
$$|m_s = -\frac{1}{2}, m_I = 0\rangle_{\text{Li}} + |m_s = +\frac{1}{2}\rangle \longrightarrow$$

$$|m_s = -\frac{1}{2}, m_I = 1\rangle_{\text{Li}} + |m_s = -\frac{1}{2}\rangle$$



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